CO#	Course outcomes
CO1	Identify key component of computer system while assembling a system.
CO2	Implement installation and configuration of computer system
CO3	Perform installation, configuration and sharing of peripheral devices.
CO4	Solve troubleshooting issues in Computer Systems
CO5	Execute dual booting.

List of assignments:

1.	Assembling and De Assembling of Computer System
2.	Loading and configuration procedure of Microsoft Client O/S Win XP /Win 7 and Windows 8
3.	Installation of utility tools (Software)
4.	Installation of utility tools (Drivers)
5.	Firewall configuration, Antivirus/Internet security loading and configuration procedure
6.	Installation and configuration of I/O devices – Printers, Webcams, Scanners.
7.	Installation and configuration of I/O devices – Digital Camera, USB Wi-fi, USB BT, USB Storages, Projectors
8.	Multiple OS loading and trouble shooting

Recommended Hardware:

All hardware component as mentioned above in the syllabus.

Text Books:

1. PC Hardware: The Complete Reference, McGraw-Hills

Reference Books:

1. The Indispensable PC Hardware Book (4th Edition) Hans-Peter Messmer
   PC Hardware: A Beginner's Guide by Ron Gilster

Course Code: UGCA1921
Course Name: Software Engineering

Program: BCA	**L:** 3 **T:**1 **P:** 0
Branch: Computer Applications	**Credits:** 4
Semester: 4th	**Contact hours:** 44 hours
Theory/Practical: Theory	**Percentage of numerical/design problems**:-
Internal max. marks: 40	**Duration of end semester exam (ESE): -**
External max. marks: 60	**Core/Elective status:** core
Total marks: 100	

Prerequisite: -
Co requisite:-

Additional material required in ESE:-

Course Outcomes: Students will be able to

CO#	**Course outcomes**
CO1	Highlight the need of software engineering
CO2	Outline the phases and activities involved in the conventional software life cycle models
CO3	Design documents for various phases of software life cycle.
CO4	Compute the complexity of the software based on multiple metrices.
CO5	Identify the tools needed for different types of documents required in software engineering.

Detailed contents	**Contact hours**
Unit 1 The Nature of Software, Need of Software Engineering, Prescriptive Process Models, Specialized Process Models, The Unified Process. [CO1]	10
Unit 2 Role of a system analyst, SRS, Properties of a good SRS document, functional and non-functional requirements, Decision tree and Decision table, Formal Requirements Specification, Software Cost Estimation. [CO2]	10
Unit 3 Software design and its activities, Preliminary and detailed design activities, Characteristics of a good software design, Features of a design document, Cohesion and Coupling, Structured Analysis, Function Oriented Design, Object-Oriented Design. [CO3]	12
Unit 4 Testing Fundamentals, Unit Testing, Integration Testing, Validation Testing, System Testing, Maintenance and Reengineering, Measures, Metrics, and Indicators, Software Measurement, Metrics for Requirements Model, Metrics for Design Model, Metrics for Testing, Metrics for Maintenance. [CO4] [CO5]	12

Text Books: